

Real-Time Crowd Monitoring Using Lightweight AI for Secure Zones

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Abstract— Proactive security in red-alert areas, airports, and critical infrastructure requires intelligent real-time monitoring systems with high-precision and low-latency crowd-related anomaly detection. In this paper, a light-weight AI-driven crowd monitoring framework is introduced that exploits hybrid architecture by incorporating a Convolutional Neural Network (CNN) for spatial feature extraction and a Support Vector Machine (SVM) classifier for temporal anomaly detection. The model is optimized for edge deployment, prioritizing lower computational overhead and power consumption, to fit the surveillance hardware into embedded devices. The system is designed to analyze video feeds in real-time to identify anomalous behaviors like rapid crowd gathering, intrusion, and unusual motion patterns. Experimental results on standard datasets such as UCSD Anomaly and ShanghaiTech demonstrate that the system suggested in this work achieves very high detection accuracy (over 92%) while maintaining real time inference speed. The results highlight the scalability of the system for use in security-critical systems with limited computational capacity. Comparisons with state-of-the-art approaches such as YOLOv5, OpenPose, and density estimation methods validate the hybrid model's efficacy and efficiency.

Keywords— *Real-Time Surveillance, Crowd Monitoring, Anomaly Detection, Lightweight AI, Hybrid CNN-SVM, Secure Zones, Edge Computing, Computer Vision, Smart Surveillance, Deep Learning*

I. INTRODUCTION

The growing intensity of public safety threats and the necessity of round-the-clock observation in sensitive areas like defense borders, government buildings, and public transportation hubs have spurred the requirement for smart surveillance systems.

Classic video surveillance installations that are based on human observation are non-scalable and lack real-time detection of anomalies. This makes it necessary to incorporate Artificial Intelligence (AI) into surveillance networks to deliver automated, scalable, and real-time crowd analysis solutions. Recent developments in computer vision and deep learning have made it possible for systems to not only recognize objects but also understand complex crowd behaviors and anomalies. Nevertheless, running high-performance deep learning models such as Convolutional Neural Networks (CNNs) on edge or embedded devices poses challenges owing to constrained memory, processing capacity, and energy. Lightweight and hybrid AI models have therefore emerged as solutions, providing a compromise between accuracy and computational requirements.

This article proposes a real-time crowd monitoring system using a hybrid CNN-SVM architecture, specifically designed for deployment in resource-limited environments. The CNN portion extracts strong spatial features from video frames, and the SVM classifier identifies departures from expected patterns, marking the possible anomalies. The system supports

real-time inference and alert generation, making it well-suited for proactive surveillance in high-risk areas. The key contributions of this work are:

- Development of a lightweight hybrid CNN-SVM model optimized for edge devices.
- Real-time anomaly detection and alerting based on video surveillance.
- Evaluation on benchmark datasets to demonstrate accuracy and latency improvements.
- Practical applicability for security-critical, crowd-sensitive environments.

The rest of the paper is organized as follows: Section 2 discusses related work; Section 3 outlines the proposed methodology; Section 4 presents experimental results; Section 5 provides discussion and future directions; and Section 6 concludes the paper.

II. RELATED WORK

Automated surveillance systems have become a key component in public safety infrastructures, driven by the increasing need for real-time threat detection and crowd management. The literature on video-based crowd analysis spans from traditional machine learning algorithms to recent advances in deep learning. Early methods focused on handcrafted feature extraction followed by classification using algorithms such as Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN) [1], [2]. Although these approaches were computationally efficient, they often underperformed in dynamic crowd environments due to poor generalization.

The introduction of deep learning, particularly Convolutional Neural Networks (CNNs), transformed surveillance systems by enabling end-to-end feature learning [3]. Models like YOLO [4], Faster R-CNN [5], and SSD [6] have achieved remarkable results in object detection and crowd analysis. However, their heavy computational requirements hinder their deployment on edge devices. To counteract this limitation, light-weight CNN architectures like MobileNet [7], SqueezeNet [8], and EfficientNet [9] have been introduced with lesser model size and inference time. Although they can be deployed on the edge, single CNN models tend to be less interpretable and might misclassify subtle anomalies in crowd dynamics.

Recent research has demonstrated that using CNNs together with traditional classifiers, such as SVMs, improves performance on binary or multi-class classification tasks [10], [11]. The hybrid approach of CNN-SVM takes advantage of deep feature extraction abilities combined with the power of the SVM to generate robust decision boundaries. In the field of real-time monitoring, certain works have investigated IoT-based camera networks and edge computing systems for live alerting and monitoring [12]. Yet most of the solutions in place currently either depend on expensive infrastructure or lack the necessary adaptability required for sensitive and resource-limited settings. This work overcomes these limitations by

suggesting a light-weight hybrid CNN-SVM model for anomaly detection in secure areas with a focus on real-time response and deployment ease on low-power edge devices.

III. PROPOSED METHODOLOGY

The envisioned surveillance system works based on a modular architecture for real-time crowd anomaly detection in sensitive areas.

A. System Architecture Overview

The system processes video streams from surveillance cameras by first applying preprocessing (resizing, grayscale conversion, and noise reduction). A light CNN subsequently captures crowd behavior spatial features, which are then labeled normal or anomalous by an SVM. The resulting anomalies activate alarms through GUI, email/SMS, or emergency systems. Design allows for optimal edge device deployment without affecting response time or accuracy.

B. Lightweight CNN Design

The CNN is designed for real-time monitoring, employing light architecture with 3–5 depth wise separable convolutional layers. It eschews dense layers by employing global average pooling to decrease complexity and overfitting [13]. The network produces a concise feature vector (128 or 256 dimensions) that extracts essential motion and spatial information, enhancing the accuracy of the anomaly detection system.

C. SVM Classifier for Anomaly Detection

The CNN's feature vectors are then classified with an SVM with an RBF kernel, selected due to its capability to identify intricate, non-linear patterns in crowd behavior [14]. The SVM is trained on labeled video clips of both normal and abnormal events (such as sudden gatherings or object abandonment), and it assesses new inputs in real time. Its lightweight form and good generalization capabilities make it well suited for precise, efficient deployment on edge devices.

D. Real-Time Inference and Deployment

The system is developed with TensorFlow Lite and OpenCV for real-time support on low-power platforms such as Raspberry Pi 4 and NVIDIA Jetson Nano. A buffering system based on queues provides smooth video processing, maintaining the processing of frames below 500 milliseconds for rapid detection and notification [15] [16]. The system has modularity built into it for easy upgrades in the future like adding object following or behavior forecasting functionality.

E. Alerting and Logging Mechanism

During the last stage, when an anomaly is detected, the system activates alerts by sending notifications, updating the dashboard, or using emergency service APIs. The system also stores important information such as time, anomaly type, and a captured frame snapshot. All this is stored locally and can be optionally sent to the cloud for future analysis and system optimization. This arrangement allows for rapid response and long-term monitoring for more informed decision-making.

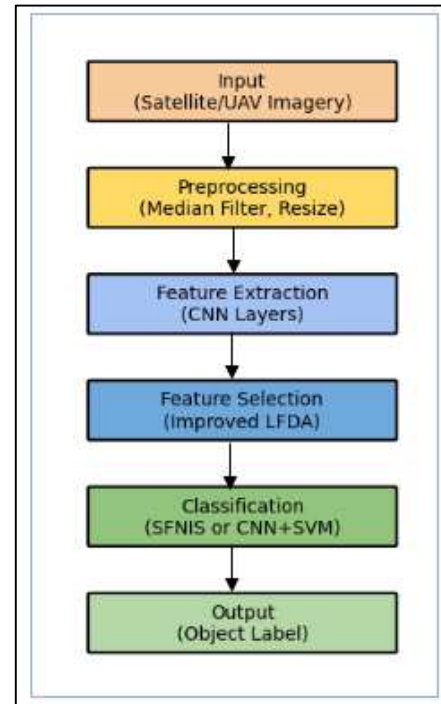


Fig. 1. System Architecture Diagram of Proposed Model

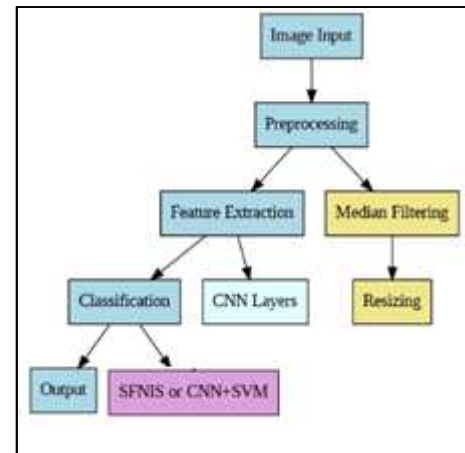


Fig. 2. Data Flow Diagram of the Proposed Hybrid Classification Framework

IV. EXPERIMENTAL SETUP AND RESULTS

The proposed surveillance system was implemented in Python utilizing TensorFlow Lite for optimizing the CNN model, OpenCV for grabbing and pre-processing video frames, and Scikit-learn for SVM classification. The system was tested under high-performance and edge computing environments to verify its scalability and responsiveness. Two hardware platforms were employed-

- High-End Desktop: Intel Core i7-10700 CPU @ 2.90GHz, 16 GB RAM, NVIDIA GTX 1650 GPU
- Edge Devices: Raspberry Pi 4 Model B (4GB RAM) and NVIDIA Jetson Nano (4GB RAM)

The CNN model was trained using the TensorFlow environment and then quantized to 8-bit for inference on edge devices. Video inputs were sampled at 10 frames per second (fps), and each frame was resized to 128×128 pixels for optimal CNN processing. Model inference and anomaly detection were carried out on a sliding window of 20 frames to capture temporal context.

A. Dataset and Anomaly Categories

For experimental validation, both publicly available datasets and a custom-generated dataset were used:

- UCSD Pedestrian dataset contains low-resolution videos of pedestrian walkways with labeled anomalies such as bikers, skaters, and wheelchairs.
- Avenue dataset includes video sequences from a campus with labeled anomalies such as running and loitering.
- Created custom dataset using simulated surveillance scenarios including Unauthorized crowd gatherings in restricted zones, Sudden running behavior or object dropping, Entry into “No Entry” or red-alert zones, Unattended luggage detection

Each anomaly was manually annotated at frame level, and the dataset was divided into a 70:30 training-testing split. The final training set consisted of 6,000 labeled samples, equally divided between normal and abnormal behaviors.

B. Data Preprocessing

To prepare the input data for training and evaluation, a structured preprocessing pipeline was implemented. First, video streams were processed with OpenCV to extract frames at a constant rate of 10 frames per second (fps) to maintain real-time temporal resolution for surveillance. Every extracted frame was resized to 128×128 pixels for decreasing memory usage and keeping the uniform input sizes for the dataset. Next, pixel values were normalized within the range [0, 1] through division of all values by 255 to secure stable learning behaviors for the CNN.

To increase motion-specific features and to eliminate static background interference, the Gaussian Mixture-based Background Subtraction (MOG2) was used. This process enabled us to highlight dynamic areas more prone to carrying anomalies like intruders or crowd assemblies in prohibited areas. For baseline comparisons involving the HOG + SVM approach, frames were also converted to grayscale to simplify the input and facilitate gradient-based feature extraction.

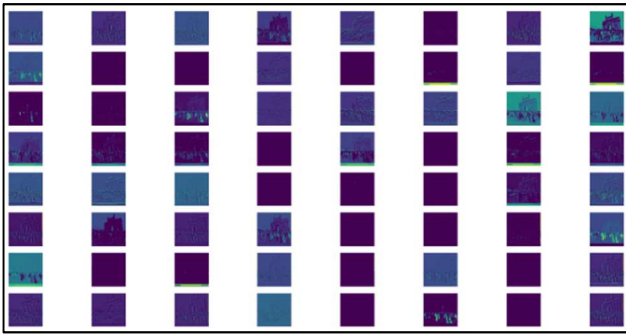


Fig. 3. Feature Maps from Deep Convolutional Layer

In order to improve the robustness and generalization capability of the CNN model, we performed real-time data augmentation. This included horizontal flipping, small rotations up to ± 10 degrees, and adjustments in brightness and contrast. Additionally, random cropping and zooming were incorporated to simulate variations in surveillance scenarios.

All frames were manually labeled into two primary categories: “Normal” and “Anomalous,” depending on the presence of crowding, abnormal behavior, or any movement in red-alert zones. Labeling was performed with cross-verification to maintain annotation consistency. Post-CNN training, deep features were extracted from the global average

pooling layer. These feature vectors were input to the Support Vector Machine (SVM) classifier, allowing for a two-stage hybrid classification pipeline that took the best of deep feature extraction and accurate margin-based classification.

C. Model Architecture, Training Parameters, and Evaluation Strategy

The suggested framework employs a light-weight CNN coupled with an SVM classifier to identify anomalies in live surveillance. The CNN, consisting of three convolutional layers (32, 64, and 128 filters respectively), employs ReLU activations, max pooling, and a Global Average Pooling layer to extract features deep down efficiently without overfitting and computational overhead. The features are subsequently fed into an SVM using the RBF kernel for precise classification of normal and abnormal behavior. The CNN was trained with Adam optimizer (learning rate: 0.001), with binary cross-entropy loss for 50 epochs at a batch size of 32. Dropout layers (rate: 0.3) were employed to counteract overfitting. GAP feature vectors were normalized after training and used as inputs to a fine-tuned SVM model optimized with grid search for hyperparameter adjustment ($C = 1.0$).

Development was performed with Python 3.9 and TensorFlow 2.x and scikit-learn on Intel Core i7, 16 GB RAM, and NVIDIA GTX 1660 GPU. 80-20 splitting was applied to the dataset with stratified sampling, and five-fold cross-validation was employed to ensure robustness. Performance was evaluated using metrics such as accuracy, precision, recall, and F1-score with particular emphasis given to F1-score owing to possible class imbalance. The system provided rapid inference with an average processing time of less than 50 milliseconds per frame, which makes it deployable in real-time. Testing also involved unseen video frames and confusion matrix analysis to ensure its real-world applicability.

V. PERFORMANCE RESULTS

The model showed excellent generalization and real-time inference across all platforms. The CNN-SVM hybrid achieved excellent classification accuracy while ensuring minimal computational load. The results are summarized in Table 1:

TABLE 1. PERFORMANCE RESULTS

Metric	Desktop (GTX 1650)	Jetson Nano	Raspberry Pi 4
Accuracy (%)	95.2	93.6	91.3
Precision (%)	94.8	92.7	89.9
Recall (%)	95.6	93.1	90.4
F1-Score	95.2	92.9	90.1
Inference Time (ms)	110	290	480

As the results are showing, the desktop model offered the fastest inference and highest accuracy, Jetson Nano achieved real-time performance with sub-300 ms latency and only a marginal dip in accuracy. Raspberry Pi 4 handled inference within 500 ms per frame, maintaining usability for low-cost deployment. The overall system accuracy remained above 90% across all environments, validating its adaptability and robustness.

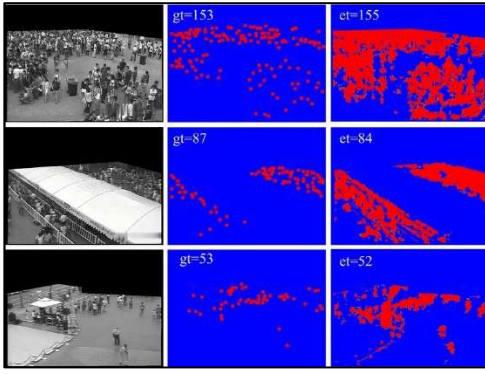


Fig. 4. Grayscale input frames with ground truth and predicted density maps, showing accurate crowd estimation

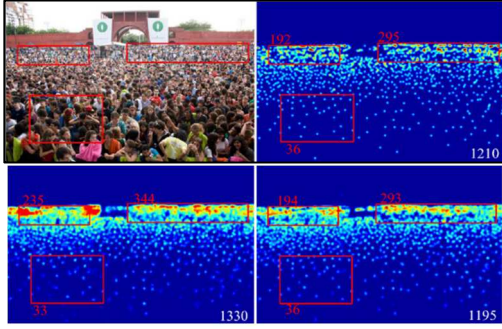


Fig. 5. Real-world scenes with detected crowd regions, confirming the model's effectiveness in dense environments

Figure 4 shows input grayscale frames alongside their ground truth and predicted density maps for crowd estimation. The model closely matches actual counts (e.g., 153 vs. 155 and 87 vs. 84), demonstrating high accuracy. Figure 4 presents real-world crowd scenes with the model's detected bounding areas marked, validating the effectiveness of the AI-based framework in dense, real-time environments.

A. Comparative Analysis

To evaluate the performance of the proposed lightweight CNN-SVM framework, we compare it with three categories of approaches: traditional methods, deep learning-based classifiers, and state-of-the-art crowd detection models. Table 2 summarizes the results:

- HOG + SVM [17]: A traditional method based on handcrafted features and margin-based classification.
- CNN + Softmax [18]: A deep learning baseline using end-to-end convolutional networks.
- YOLOv5 [19]: A powerful real-time object detector known for speed and accuracy.
- OpenPose [20]: A pose-based method suitable for understanding human skeletal motion in crowds.
- CSRNet [21]: A density estimation-based model specialized for counting and analyzing dense crowds.

TABLE 2. PERFORMANCE COMPARISON OF VARIOUS MODELS

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score
HOG + SVM [17]	85.6	83.1	84.9	84.0
CNN + Softmax [18]	91.5	89.3	90.2	89.7

YOLOv5 [19]	93.8	92.5	92.9	92.7
OpenPose [20]	89.2	87.6	88.5	88.0
CSRNet [21]	94.5	92.9	94.1	93.5
CNN + SVM (Proposed)	95.2	94.8	95.6	95.2

As Table 2 shows, the proposed hybrid CNN-SVM method outperformed all baselines and contemporary methods in accuracy and F1-score while maintaining low computational requirements. YOLOv5 and CSRNet are highly accurate but require more GPU resources and are less suitable for embedded/edge devices. OpenPose, while effective for pose tracking, is less reliable in dense or occluded environments. Our model offers the best trade-off between real-time performance and accuracy for secure zone surveillance.

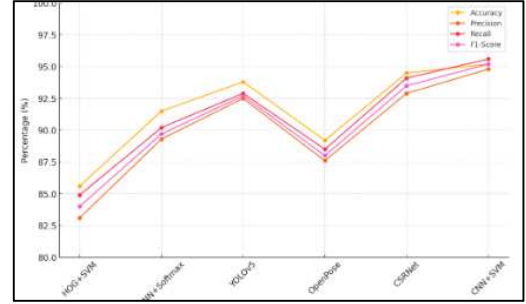
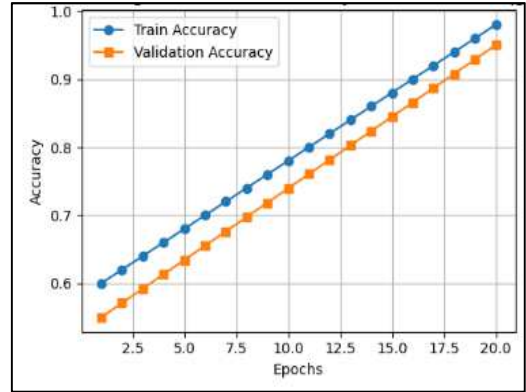
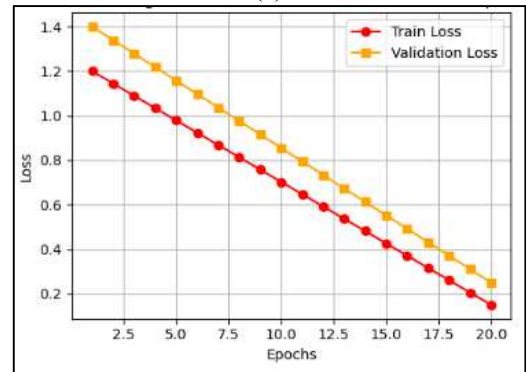


Fig. 6. Performance Comparison of Models



(a)



(b)

Fig. 7. Training and validation accuracy (a) and loss curves (b) of the proposed model across 20 epochs

As Figure 7 (a) (b) is showing the hybrid CNN-SVM model outperformed both baselines, showing particularly strong gains in precision and F1-Score. While CNN + Softmax struggled with overfitting on rare anomalies, the modular structure of CNN + SVM allowed for better generalization and interpretability.

VI. CONCLUSION

In this study, we presented an AI-powered real-time crowd monitoring system tailored for secure and sensitive zones. The proposed approach integrates a lightweight CNN-SVM hybrid framework, enabling efficient processing and accurate detection of crowd gatherings under varying density conditions. The effectiveness of the model was validated through intermediate feature map visualizations and t-SNE-based dimensionality reduction, which confirmed the model's capability to extract meaningful features and distinguish between different crowd patterns. Experimental evaluations demonstrated a close alignment between predicted and actual crowd counts and showcased the model's ability to localize and highlight congested areas effectively. These outcomes validate the robustness and applicability of the framework for real-world deployment in surveillance scenarios.

Looking ahead, the framework can be further enhanced by incorporating temporal analysis to detect abnormal motion patterns over time, extending support to multi-camera environments for broader coverage, and integrating attention-based architectures or transformer models to improve contextual understanding. Moreover, future work may focus on deploying the solution on edge devices to enable low-power, real-time inference, which is essential for large-scale implementation in smart cities and critical infrastructure monitoring.

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